Use scale factors

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: A length of 7 cm is enlarged by scale factor 3.5. What is the new length? Copy or complete the first step.

2. A recipe for 6 uses 450 g flour. How much flour is needed for 14?

3. In the ratio 4:7, the smaller part is 20. What is the larger part?

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A length of 7 cm is enlarged by scale factor 3.5. What is the new length? Copy or complete the first step.
   Answer 24.5 cm.

2. A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer 1,050 g.

3. In the ratio 4:7, the smaller part is 20. What is the larger part?
   Answer 35.

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

Use scale factors

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

2. A recipe for 6 uses 450 g flour. How much flour is needed for 14?

3. In the ratio 4:7, the smaller part is 20. What is the larger part?

4. Solve this independently and show every stage: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

5. Use a second method or representation for: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

6. Check this result using an inverse, estimate or known fact: In the ratio 4:7, the smaller part is 20. What is the larger part?

7. Prove or explain your reasoning: Share £420 in the ratio 3:4.

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: In the ratio 4:7, the smaller part is 20. What is the larger part?

Teacher answers and checking prompts

1. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer 24.5 cm.

2. A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer 1,050 g.

3. In the ratio 4:7, the smaller part is 20. What is the larger part?
   Answer 35.

4. Solve this independently and show every stage: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer 24.5 cm.

5. Use a second method or representation for: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer Expected result: 1,050 g. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: In the ratio 4:7, the smaller part is 20. What is the larger part?
   Answer Expected result: 35. A valid check must be shown.

7. Prove or explain your reasoning: Share £420 in the ratio 3:4.
   Answer £180 and £240. The explanation must show why the method works.

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: In the ratio 4:7, the smaller part is 20. What is the larger part?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use scale factors

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Share £420 in the ratio 3:4.

2. Prove the answer to this question, rather than only calculating it: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

3. Solve this in two different ways and compare them: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

4. Work backwards from the answer to reconstruct the question: 35.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.

6. Create a new example that tests the same idea as: In the ratio 4:7, the smaller part is 20. What is the larger part?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Share £420 in the ratio 3:4.
   Answer £180 and £240. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer Expected result: 24.5 cm. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer Expected result: 1,050 g. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 35.
   Answer One valid reconstruction is: In the ratio 4:7, the smaller part is 20. What is the larger part? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

6. Create a new example that tests the same idea as: In the ratio 4:7, the smaller part is 20. What is the larger part?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.